# Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Fairfax, Virginia

## PROFESSIONAL SUMMARY

Python software engineer with 7+ years building production systems in regulated industries - 7 years at Entrata (multifamily/affordable housing SaaS: HUD/LIHTC compliance, PHI-protected data, 100K+ monthly active users). Currently at Intuit integrating Anthropic Claude into financial applications: Claude API pipelines, prompt chains, structured outputs, LLM evaluation frameworks, and compliance guardrails. Built Leasel, an 8-agent LangGraph HITL platform for lease compliance with audit-ready evidence trails, and AgentProbe, an open-source LLM evaluation toolkit. Familiar with Claude Code, Anthropic APIs, and MCP. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Portfolio: venkatalakshmiparimala.github.io/Portfolio

## CORE COMPETENCIES

Claude API / Anthropic | Claude Code | MCP (Model Context Protocol) | Agentic Workflows | Human-in-the-Loop (HITL) | RAG Pipelines | Structured Outputs | LLM Evaluation & Governance | Prompt Engineering | Compliance & Regulated AI | Python / FastAPI | PostgreSQL / REST APIs | AWS | LangGraph / LangChain | Real Estate Technology

## EXPERIENCE

### Intuit

Jun 2026 - Present

**Software Engineer · Virginia, USA**

- Integrate **Anthropic Claude** into Python/FastAPI services via Claude API - financial data summaries, natural-language queries, and reporting; use **Claude Code** as the primary development interface.
- Design **prompt chains and structured-output schemas** so Claude responses stay consistent and validate cleanly in downstream financial workflows.
- Build **MCP-compatible tool integrations** for structured data access and service orchestration across enterprise systems.
- Ground Claude responses in customer financial data via retrieval so every answer traces back to a source record - full auditability.
- Build **LLM evaluation and regression tests** (LLM-as-judge + golden datasets) to catch hallucinations and quality drops before release; add guardrails for input validation, sensitive-data handling, and model output fallbacks.
- Build ML pipelines (scikit-learn, XGBoost) from feature engineering to prediction APIs; containerize and ship via GitHub Actions CI/CD to AWS with Prometheus/Grafana monitoring.

### Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 - Jul 2024

**Senior Software Engineer · Regulated domain: HUD, LIHTC, PHI · 100K+ MAU, 1M+ property records**

- Delivered 20+ production features (Python, PHP, Java, React, PostgreSQL) across affordable housing and property management modules in a **regulated, compliance-first environment** - 120+ Agile sprints.
- Owned the annual **HUD income-limit data pipeline** for 6 years: transformed federal data into ~180,000 tenant eligibility records with LIHTC/HUD validation, rollback checks, and zero compliance violations.
- Built a **RabbitMQ async processing system** for rent updates across thousands of tenant records - idempotency and retry logic eliminated double-processing in a regulated financial workflow.
- Implemented **Libsodium-encrypted pipelines** for AWS S3 document migration covering 100% of resident PHI - zero security incidents; updated 50%+ of product codebase for encryption compliance.
- Optimized 200+ SQL queries (30% avg performance gain); wrote Python/Bash automation that cut manual operational work by 80%.
- Led Docker microservices redesign (50% better scalability, 30% faster deployments); reviewed 300+ pull requests, mentored junior engineers.

### Entrata India

Sep 2018 - Oct 2020

**Junior Software Engineer · Pune, India**

- Built property management and affordable housing modules (Python, PHP, Java, PostgreSQL) - leasing, income certification, and compliance workflows.
- Delivered large-scale data import and reconciliation pipelines for **1.6M HUD/Tax Credit records** with output validation and quality gates.
- Led AWS S3 cloud migration from legacy server storage; created 56 state-specific housing certification documents.

## PROJECTS

### Leasel - Multi-Agent AI Platform for Regulated Compliance

HITL · Audit Trails · Regulated AI

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · HUD API · OCR - github.com/venkatalakshmiparimala/Lease_complaince_rag

Built an AI-powered lease compliance and support investigation platform for affordable housing operations. Core design rule enforced in code: LLM extracts and reasons; deterministic or human logic makes every eligibility, compliance, and financial call - no AI auto-resolution of regulated decisions. Human-in-the-loop escalation triggers when confidence drops below threshold. Full audit trail: every answer cited to its source document clause. LLM-as-judge CI: 93.6% precision and 87.9% F1 across 1,240 test cases; 100% citation correctness on document Q&A.

### AgentProbe - Open-Source LLM Evaluation Toolkit

Claude API · Governance · Evals

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic Claude · OpenAI · Azure OpenAI · Groq · Ollama - github.com/venkatalakshmiparimala/AgentProbe

Built a 15-endpoint FastAPI backend routing prompts across five LLM providers including Anthropic Claude API. Scores LLM and RAG outputs on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance - LLM-as-judge with golden dataset CI that blocks failing runs before production. Three LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) with inter-agent state handoffs.

### AI on FHIR - Clinical NLP & LLM API Integration

HIPAA · Regulated · REST APIs

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs - github.com/venkatalakshmiparimala/AI_on_FHIR

Translates English/Spanish questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut manual clinical lookup time by ~60%. HIPAA-compliant role-based access control, sensitive-data routing, and Docker containerization. Demonstrates AI integration with regulated enterprise APIs.

## EDUCATION

**M.S. Data Analytics Engineering - George Mason University, Virginia** May 2026

GPA 3.87 · ML systems, NLP, data pipelines, statistical modeling, AI engineering

**B.Tech Computer Science & Engineering - JNTU Anantapur** 2017

## TECHNICAL SKILLS

**AI & LLMs:** Anthropic Claude (API + Claude Code), MCP, LangGraph, LangChain, RAG pipelines, embeddings, vector search, prompt engineering, structured outputs, function/tool calling, HITL, LLM-as-judge evaluation, AI governance, ChromaDB, Pinecone, Weaviate, Elasticsearch, BERT, spaCy, HuggingFace

**Python & Backend:** Python (9+ yrs), FastAPI, Flask, Django, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, REST APIs, GraphQL, microservices, OAuth, HIPAA/PHI compliance

**Cloud & DevOps:** AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Prometheus, Grafana, OpenTelemetry, Bash

**Data & ML:** Pandas, NumPy, scikit-learn, XGBoost, PyTorch, PySpark, ETL pipelines, Snowflake, data modeling, query optimization, Azure Data Factory, Databricks

**Other Languages:** Java, Spring Boot, TypeScript, JavaScript, Node.js, React, Redux, PHP, SQL, Scala **Messaging:** RabbitMQ, Kafka, Celery